

Lecture 11

Deep Learning (Agent Perspective)

COMP532 – Machine Learning and Bioinspired Optimisation

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Why Another Deep Learning Lecture?

Deep learning appears in many courses.

In this course, we study deep learning as:

- ▶ A representation engine
- ▶ A function approximator
- ▶ The computational substrate of intelligent agents

Roadmap

1. Intelligent agents and representation
2. Neural networks as function approximators
3. Depth and expressivity
4. Inductive bias
5. Deep learning inside agents

What is an Intelligent Agent?

Agent-environment interaction:

$$s_t \rightarrow a_t \rightarrow r_{t+1} \rightarrow s_{t+1}$$

Core components:

- ▶ Perception
- ▶ Policy
- ▶ Value estimation
- ▶ World model
- ▶ Memory

How do we represent these components?

The Representation Problem

Intelligence requires approximating unknown functions:

$$f^*(x)$$

Examples:

- ▶ Image $\rightarrow$ object label
- ▶ State $\rightarrow$ action
- ▶ State $\rightarrow$ value
- ▶ Token sequence $\rightarrow$ next token

Function Approximation

We approximate unknown functions with parameterized models:

$$f_{\theta}(x)$$

where θ are learnable parameters.

Learning:

$$\min_{\theta} \mathcal{L}(f_{\theta}(x), y)$$

Why Not Lookup Tables?

Tabular methods store:

$$Q(s, a)$$

For large state spaces:

- ▶ Images
- ▶ Continuous control
- ▶ Language sequences

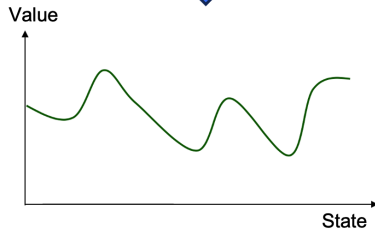
State space becomes exponential.

We need parameterized approximators.

Q-table

	A	B	C
1		-5	+10
2		-5	
3	A1/B3	+1	-1

State	Left	Right	Down	Up
A1	-	-5	0	-
A2	0	0	0	0
B1	0	0	0	0
B2	0	0	0	0
B3	0	0	0	0
C1	0	0	0	0
C2	0	0	0	0
C3	0	0	0	0



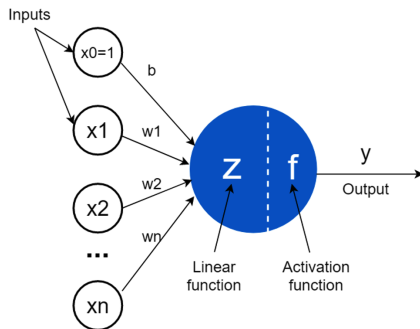
A Single Neuron

Basic building block:

$$y = \sigma(w^T x + b)$$

where:

- ▶ $x \in \mathbb{R}^d$
- ▶ $w \in \mathbb{R}^d$
- ▶ $b \in \mathbb{R}$
- ▶ σ is a nonlinearity (activation function)



Why Nonlinearity Matters

Without nonlinearity:

Multiple layers collapse into a single linear transformation.

Nonlinearity enables:

- ▶ Complex decision boundaries
- ▶ Hierarchical representations
- ▶ Compositional structure

Deep Networks as Function Composition

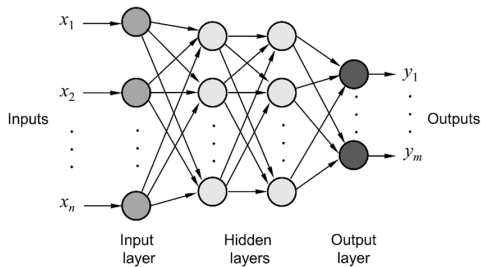
Layered computation:

$$h_1 = \sigma(W_1 x)$$

$$h_2 = \sigma(W_2 h_1)$$

$$f_{\theta}(x) = W_L h_{L-1}$$

Deep learning is composition of nonlinear functions.



Hierarchical Abstraction

Examples:

Vision:

pixels $\rightarrow$ edges $\rightarrow$ shapes $\rightarrow$ objects

Language:

tokens $\rightarrow$ phrases $\rightarrow$ semantics

Universal Approximation (Informal)

Theorem (informal):

A sufficiently wide neural network can approximate any continuous function on compact domains.

However:

- ▶ Width is inefficient
- ▶ Depth enables compositional efficiency

Depth vs Width

Consider compositional functions:

$$f(x) = f_1(f_2(f_3(x)))$$

Deep networks naturally represent such structure.

Shallow networks may require exponentially many neurons.

What is Inductive Bias?

Inductive bias = structural assumptions about the problem.

Examples:

- ▶ CNN: translation invariance
- ▶ RNN: sequential structure
- ▶ Transformer: relational attention

Bias vs Data

Two regimes:

- ▶ Strong inductive bias, limited data
- ▶ Weak bias, massive data (foundation models)

Modern large models rely more on scale.

Perception Modules

Deep networks map observations to representations:

$$O_t \rightarrow S_t$$

Examples:

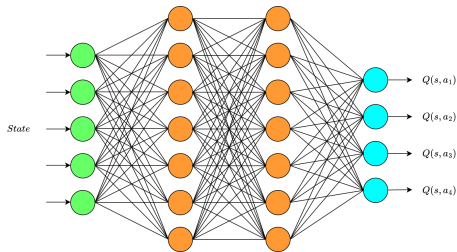
- ▶ CNN embeddings
- ▶ Transformer embeddings

Value Function Approximation

In reinforcement learning:

$$Q_{\theta}(s, a)$$

Deep networks approximate value functions.

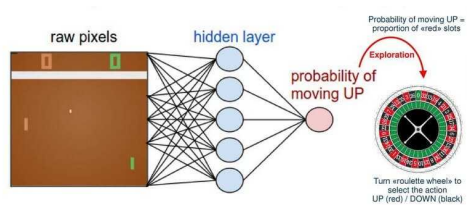


Policy Approximation

Parameterized policies:

$$\pi_{\theta}(a|s)$$

Neural networks directly output action probabilities.



World Models

Model-based learning:

$$p_{\theta}(s_{t+1}|s_t, a_t)$$

Large language models:

$$p_{\theta}(x_{t+1}|x_{\leq t})$$

Deep models approximate environment dynamics.

Deep Learning as Agent Substrate

Neural networks can represent:

- ▶ Perception
- ▶ Policy
- ▶ Value
- ▶ World model
- ▶ Memory (attention)

Learning as Optimization

Learning solves:

$$\min_{\theta} \mathcal{L}(f_{\theta}(x), y)$$

Optimization drives representation emergence.

Next lecture:

- ▶ Gradient descent
- ▶ Credit assignment
- ▶ Training dynamics

Key Takeaways

1. Deep learning is function approximation.
2. Depth enables compositional representation.
3. Inductive bias controls generalization.
4. Neural networks are computational cores of agents.